

decoding the RC-5 command on pressing the required keys of the

remote control and providing high logic to enable corresponding bilateral

TABLE I
Pin Details of CCTV Camera DIN Socket

Socket	Camera	Signal	Switch (IC)	Enable pins to microcontroller
SK1	Camera-1	Video	S2a (IC4)	Port (RC7)-pin 18
SK1	Camera-1	Audio	S2b (IC4)	Port (RC7)-pin 18
SK2	Camera-2	Video	S3a (IC4)	Port (RC6)-pin 17
SK2	Camera-2	Audio	S3b (IC4)	Port (RC6)-pin 17
SK3	Camera-3	Video	S4a (IC3)	Port (RC5)-pin 16
SK3	Camera-3	Audio	S4b (IC3)	Port (RC5)-pin 16
SK4	Camera-4	Video	S5a (IC3)	Port (RC4)-pin 15
SK4	Camera-4	Audio	S5b (IC3)	Port (RC4)-pin 15
SK5	Camera-5	Video	S6a (IC2)	Port (RC1)-pin 12
SK5	Camera-5	Audio	S6b (IC2)	Port (RC1)-pin 12
SK6	Camera-6	Video	S7a (IC2)	Port (RC0)-pin 11
SK6	Camera-6	Audio	S7b (IC2)	Port (RC0)-pin 11

TABLE II
Truth Table for Port C

PORT	RC7	RC6	RC5	RC4	RC1	RC0	Camera-ON
Port C	1	0	0	0	0	0	Camera-1
Port C	0	1	0	0	0	0	Camera-2
Port C	0	0	1	0	0	0	Camera-3
Port C	0	0	0	1	0	0	Camera-4
Port C	0	0	0	0	1	0	Camera-5
Port C	0	0	0	0	0	1	Camera-6

A remote sensor is connected to port RB0 (pin 21), called interrupt port, of the controller and the software initialises interrupt routines along with timer-0 interrupt to decode the RC-5 command. The RC-5 protocol was described in March 2007 issue of EFY, so it is not discussed here.

Any transmission from the remote consists of two start bits, one toggle bit, 5-bit address and 6-bit command. Here we have not used the 5-bit address. A biphasic-modulated bit can be thought of as two separate bits that are always the inverse of each other. A logical zero is represented by a '10' pattern on the IR input, while a logical one is represented by a '01' pattern. That is basically used to decode the received message.

Toggle bit is a particular property of the RC-5 protocol. This bit changes polarity every time you press a key and will remain unchanged as long as you hold the key. That enables the receiver to detect released keys, which helps to eliminate key bounces. We have also used this bit to detect so that if any key is kept pressed for a long time, the routine detects the command once, keeps the value in a general-purpose register called 'passdummy' (refer to '.asm' file), and does not process and store any further command till the remote key is released. The stored RC-5 command value in register 'passdummy' is used to perform the necessary function given in the source code.

The LCD shows the camera number selected from the remote and mode of operation (auto/manual).

For camera-hold time set for auto mode and other display text information, please see the LCD screenshot in Fig. 3. Use of the LCD makes the project more user-friendly.

All the cameras are to be connected to SK1 through SK6 (5-pin DIN connector) as shown in the circuit diagram. The extreme right and left pins are for 12V positive and negative supply (to be fed from the regulated power supply of the main switcher)

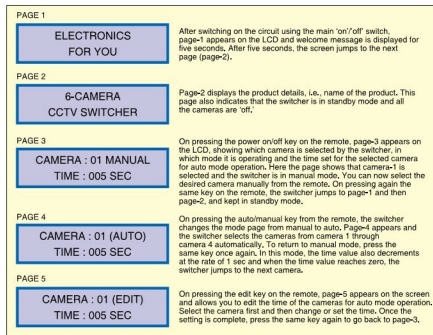


Fig. 3: Screenshots of LCD